

Matthew C. Voynovich

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EDUCATION

RENSSELAER POLYTECHNIC INSTITUTE

M.S. in Computer Science

Concentration in AI, Machine Learning, and Data Science

Troy, NY

Jan 2026 - May 2027

RENSSELAER POLYTECHNIC INSTITUTE

Dual Major B.S. in Computer Science and B.S. Information Technology & Web Science

Concentration in AI, Machine Learning, and Data Science

Troy, NY

Aug 2023 - May 2026

Cumulative GPA: 3.97/4.0; Dean's List 2023-2025; Accelerated to Graduate in 3 years

Relevant Coursework: Managing IT Resources, Algorithms, Database Systems, Web Science Development, Computational Vision, Intro to Logic Based AI, Machine Learning from Data, Data Science, Decentralized AI, Projects in AI & ML

Awards and Leadership: 5x Dean's Honor List; President of the Rensselaer Running Club, Member of RPAI club, UPE Honor Society

WORK EXPERIENCE

FULL STACK DEVELOPER

RPI, NY

Contract Work for Novella Prep

June 2025 – Aug 2025

- Spearheaded the development of a new product line that opened an entirely new revenue stream for a test prep company.
- Led development of Zeewa, an adaptive learning platform featuring diagnostic exams, AI-generated practice questions, interactive lessons, and assignment workflows.
- Built a scalable Vue.js + Node.js + PostgreSQL (Supabase) system using TypeScript, Docker, and Nginx for secure multi-tenant deployment. Integrated OpenAI LLMs for dynamic content creation and implemented real-time LaTeX/Markdown rendering, SSE pipelines, and stateful authentication. Managed full deployment workflow with DigitalOcean, Render, and Porkbun

RESEARCH

AI AGENT SYSTEMS & WORKFLOW SAFETY

RPI, NY

Graduate Researcher, RPI

Jan 2025 – Present

- Conducting research on AI agents' automatic workflow generation to ensure compliance with privacy and security policies.
- Studying agent planning, tool usage, and data flow in frameworks such as LangChain and LangGraph.
- Developing techniques to analyze and mitigate sensitive information leakage in agent-based systems.

OPTIMIZATION IN QUANTUM

RPI, NY

Undergraduate Researcher, RPI

Jun 2025 – Dec 2025

- Conducted research on the applications of quantum computing in optimization problems, implementing the Quantum Approximate Optimization Algorithm (QAOA), comparing quantum techniques to classical and machine learning-based techniques.
- Used hybrid-classical optimization techniques in JAX and Qiskit to solve seismic inversion problems.

INVESTIGATIONS OF QUANTUM PHASE ESTIMATION

RPI, NY

Undergraduate Researcher, RPI

Jan 2025 – Jun 2025

- Researching applications of Quantum Phase Estimation (QPE) in Quantum Signal Processing. Exploring the mathematical foundations of quantum Fourier transforms, eigenvalue estimation, and phase estimation techniques to enhance signal processing.
- Designing, implementing, and optimizing quantum circuits using Qiskit, leveraging IBM's quantum computing framework to simulate and test QPE-based algorithms and testing said quantum programs on the RPI Quantum Computer.

PROJECTS

DECENTRALIZED MULTI-AGENT RL TRAFFIC CONTROL

RPI, NY

Developer

Apr 2026 – May 2026

- Developed a decentralized multi-agent reinforcement learning traffic control system in PyTorch using Deep Q-Networks (DQN) and a parameter-sharing CTDE (Centralized Training with Decentralized Execution) architecture to optimize traffic flow and stabilize scalable learning across 27 intersections in downtown Troy, NY.
- Achieved significant performance improvements over traditional traffic control strategies, reducing average vehicle waiting time from ~36 seconds to ~2 seconds in simulation and reducing queue lengths by ~6x.

ENTERPRISE SEARCH AGENT

RPI, NY

Independent Contractor

Feb 2026 – May 2026

- Built an AI-powered enterprise search prototype for Deloitte as part of a capstone project using Next.js (TypeScript), FastAPI, and PostgreSQL; implemented document ingestion, chunking, and embedding generation pipelines, then combined semantic vector search (pgvector) with BM25 keyword retrieval (ParadeDB) and reranking to deliver more relevant natural-language results.
- Designed the platform as an open-source, self-hosted solution to keep sensitive enterprise data fully onsite, reducing third-party exposure and strengthening security/compliance posture.

OREGANO THYME

RPI, NY

Independent Contractor, Full Stack Developer

Sep 2025 – Nov 2025

- Collaborated in a small team to develop a full-stack website for a local Troy, NY restaurant with limited online presence, leading backend development and integrating the site with the restaurant's existing Clover POS system, while implementing React-based frontend-backend communication.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript

Web: HTML/CSS, React, Vue, Node.js, Express

Databases: SQL, MongoDB

ML/AI: Scikit-learn, PyTorch